

**Advanced Database
Management Systems
4th Semester ,
2nd Year 24F-BSCS**



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Roll# 24F-CS-057

Batch: 24F

**Subject: Advanced Database Management
Systems**

Teacher: Poonam Kewal Mal

Section: A-2 / B

DEPARTMENT OF COMPUTER SCIENCE

Lab 8

Create Required Tables:

Language

SQL

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Rows

10

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Clear Command

Find Tables

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```
CREATE TABLE MEMBERS (  
    MEMBER_ID NUMBER PRIMARY KEY,  
    NAME VARCHAR2(100),  
    CITY VARCHAR2(100)  
);
```

Results

Explain

Describe

Saved SQL

History

Table created.

0.04 seconds

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 en

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Language SQL ? Rows 10 ? Clear Command Find Tables

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```
1 CREATE TABLE BOOKS (  
2     BOOK_ID NUMBER PRIMARY KEY,  
3     TITLE VARCHAR2(150),  
4     CATEGORY VARCHAR2(100),  
5     PRICE NUMBER  
6 );  
7
```

Results Explain Describe Saved SQL History

Table created.

0.05 seconds



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Language SQL



Rows 10



Clear Command

Find Tables

A::

```
1 CREATE TABLE BORROW (  
2     BORROW_ID NUMBER PRIMARY KEY,  
3     MEMBER_ID NUMBER,  
4     BOOK_ID NUMBER,  
5     BORROW_DATE DATE,  
6     CONSTRAINT fk_member FOREIGN KEY (MEMBER_ID) REFERENCES MEMBERS(MEMBER_ID),  
7     CONSTRAINT fk_book FOREIGN KEY (BOOK_ID) REFERENCES BOOKS(BOOK_ID)  
8 );
```

Results

Explain

Describe

Saved SQL

History

Table created.

0.08 seconds

Inserting Sample Data:

```
1 INSERT INTO MEMBERS VALUES (1, 'Ali Khan', 'Karachi');
2 INSERT INTO MEMBERS VALUES (2, 'Sara Ahmed', 'Lahore');
3 INSERT INTO MEMBERS VALUES (3, 'Usman Tariq', 'Karachi');
4 INSERT INTO MEMBERS VALUES (4, 'Usama shariq', 'Karachi');
5 INSERT INTO BOOKS VALUES (101, 'Database Systems', 'Education', 500);
6 INSERT INTO BOOKS VALUES (102, 'Python Basics', 'Programming', 300);
7 INSERT INTO BOOKS VALUES (103, 'AI Fundamentals', 'Technology', 700);
8 INSERT INTO BORROW VALUES (1, 1, 101, SYSDATE-5);
9 INSERT INTO BORROW VALUES (2, 1, 102, SYSDATE-3);
10 INSERT INTO BORROW VALUES (3, 2, 103, SYSDATE-2);
11 INSERT INTO BORROW VALUES (4, 3, 101, SYSDATE-1);
12 INSERT INTO BORROW VALUES (5, 2, 102, SYSDATE-1);
13
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```

Results	Explain	Describe	Saved SQL	History
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```
1 row(s) inserted.
```

0.01 seconds

Apply JOINS:

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```
SELECT m.name, b.title, br.borrow_date
FROM borrow br
JOIN members m ON br.member_id = m.member_id
JOIN books b ON br.book_id = b.book_id;
```

Results

ExplainDescribeSaved SQLHistory

NAME	TITLE	BORROW_DATE
Ali Khan	Python Basics	4/12/2026
Ali Khan	Database Systems	4/10/2026
Sara Ahmed	Python Basics	4/14/2026
Sara Ahmed	AI Fundamentals	4/13/2026
Usman Tariq	Database Systems	4/14/2026

5 rows returned in 0.14 secondsDownload

Use Aggregate Functions:

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```
SELECT m.name, COUNT(*) AS total_books
FROM borrow br
JOIN members m ON br.member_id = m.member_id
GROUP BY m.name;
```

Results

ExplainDescribeSaved SQLHistory

NAME	TOTAL_BOOKS
Ali Khan	2
Sara Ahmed	2
Usman Tariq	1

3 rows returned in 0.15 secondsDownload

Apply Filters:

```
29
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31 SELECT m.name, m.city, COUNT(*) AS total_books
32 FROM borrow br
33 JOIN members m ON br.member_id = m.member_id
34 WHERE m.city = 'Karachi'
35 GROUP BY m.name, m.city;
36
37
38
39
40
41
42
```

Results Explain Describe Saved SQL History

NAME	CITY	TOTAL_BOOKS
Ali Khan	Karachi	2
Usman Tariq	Karachi	1

2 rows returned in 0.04 seconds Download

Computed Column:

```
36
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41 SELECT title, price,
42 price * 0.9 AS discounted_price
43 FROM books;
44
```

Results Explain Describe Saved SQL History

TITLE	PRICE	DISCOUNTED_PRICE
Database Systems	500	450
Python Basics	300	270
AI Fundamentals	700	630

3 rows returned in 0.02 seconds Download

Advanced Report (Combined Query):

```
58
59
60 SELECT m.name,
61 COUNT(br.borrow_id) AS total_books,
62 AVG(b.price) AS avg_price
63 FROM borrow br
64 JOIN members m ON br.member_id = m.member_id
65 JOIN books b ON br.book_id = b.book_id
66 GROUP BY m.name;
```

Results Explain Describe Saved SQL History

NAME	TOTAL_BOOKS	AVG_PRICE
Ali Khan	2	400
Sara Ahmed	2	500
Usman Tariq	1	500

3 rows returned in 0.00 seconds Download

Interactive Report

Q

Go

Actions

Name	Total Books	Avg Price
Ali Khan	2	400
Sara Ahmed	2	500
Usman Tariq	1	500
		1 - 3